UDAY KUMAR

Electronics & Communication Engineering • RF & Microwave Research Intern • IEEE Student Branch President

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PROFESSIONAL SUMMARY

Driven ECE undergraduate actively conducting RF & Microwave Antenna research at IIT Delhi, with dual published research papers in antenna engineering. Demonstrated expertise spans antenna simulation (CST Studio), embedded systems design (Arduino, Raspberry Pi Pico), PCB design (Altium Designer), and IoT product development. Currently architecting two product-grade systems — a palm biometric fintech platform and an IoT-driven accident detection system. Elected IEEE Student Branch President; committed to advancing research-driven innovation at the intersection of wireless communication, embedded systems, and smart electronics.

EDUCATION

B.Tech – Electronics & Communication Engineering
Manav Rachna International Institute of Research and Studies
2024 – 2028 | Delhi NCR, India
SGPA: S2 – 8.10 | S3 – 8.25 | S4 – 8.73 (Consistent Academic Excellence)

LEADERSHIP & ACTIVITIES

IEEE Student Branch President
Elected to lead university IEEE chapter; driving technical workshops, industry connections, and research culture among 100+ student members.
Active IEEE Member | Technical Hackathon Participant

RESEARCH & INTERNSHIP EXPERIENCE

RF & Microwave Antenna Research Intern | Indian Institute of Technology Delhi (IIT Delhi)
6 Weeks (Ongoing) | New Delhi, India

- Designing and simulating high-gain slotted waveguide and microstrip patch antenna arrays for X-Band RF applications using CST Studio Suite, achieving measurable improvements in radiation pattern uniformity.
- Conducting full-cycle antenna performance analysis — S-parameters, gain optimization, return loss, and bandwidth characterization — under direct supervision of IIT Delhi faculty researchers.
- Contributing to active research investigations in wireless communication systems, with outcomes feeding into co-authored research publications.

PCB Design Intern | Manav Rachna International Institute of Research and Studies
1 Month | Faridabad, India

- Executed end-to-end PCB development lifecycle — schematic capture, component placement, high-speed routing, and Design Rule Check (DRC) verification — using Altium Designer.
- Delivered production-ready PCB layouts for embedded hardware modules, ensuring signal integrity and manufacturability compliance.

CONFERENCE PAPER (ACCEPTED)

1. **IoT and Sensor Fusion in Smart Helmets: A Review Toward Safer and Sustainable Mobility**
   Accepted for Presentation • Conference hosted by VIT Bhopal University | IoT • Sensor Fusion • Smart Safety Systems • Sustainable Mobility

RESEARCH MANUSCRIPTS

Independent research manuscripts currently under development; not yet submitted for formal publication.

1. **Design and Comparative Analysis of 4-Slot and 8-Slot Slotted Waveguide Antenna Arrays in X-Band**
   Antenna Array Design • X-Band RF Systems • Gain Optimization • Radiation Pattern Analysis • RF Performance Evaluation

2. **Design and Performance Analysis of a Slot-Loaded Microstrip Patch Antenna for Enhanced Bandwidth**
   Bandwidth Enhancement • Microstrip Patch Antennas • RF Simulation • Antenna Performance Characterization

KEY PROJECTS

PixiePlot – AI-Powered Storytelling Web App *[Ongoing]*
Developing an AI-powered Progressive Web App (PWA) using Flutter for generating personalized children's stories, featuring responsive UI/UX across desktop and mobile, user authentication, and a personalized story library. Implementing voice storytelling workflow and currently integrating AI-driven story generation and voice narration. Using GitHub for version control and collaborative development, with emphasis on performance optimization, accessibility, and scalable architecture.

Smart Irrigation System *[Completed]*
Sensor-driven autonomous irrigation controller on Raspberry Pi Pico, integrating soil moisture and environmental sensors for precision water management and remote agricultural monitoring.

Binary Calculator using Arduino *[Completed]*
Embedded hardware implementation of binary arithmetic operations using Arduino; designed custom I/O interfacing and state-machine logic for real-time computation display.

Smart Helmet – Accident Detection & Alert System *[Ongoing]*
IoT safety system on Raspberry Pi Pico integrating MPU6050 IMU, GPS, and GSM modules for real-time accident detection, geolocation tagging, and automated emergency alert dispatch to first responders.

Air Quality Monitoring System *[Completed]*
Real-time environmental monitoring platform using multi-sensor gas detection arrays for continuous air quality analytics, with threshold-based alerting for particulate and gaseous pollutants.

TECHNICAL SKILLS

Languages: C, C++, Python, HTML, CSS, JavaScript

Embedded Systems: Arduino IDE, Raspberry Pi Pico, Microcontrollers, Sensor Interfacing

EDA & Simulation: Altium Designer, Cadence, CST Studio Suite

IoT & Hardware: IoT Prototyping, Hardware Design, Electronics Integration

Research Areas: RF/Microwave Engineering, Antenna Design, Wireless Communication, FinTech

OS & Tools: Linux, Git, Embedded Debugging